

Public Credit Opportunity Engine — Interview Prep Guide

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Public Credit Opportunity Engine

Interview Preparation Guide — Visual Edition

"Where is compensation in public credit markets right now, and what is the expected payoff of trading volatility in those segments?"

System Architecture at a Glance



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 Data Flow: US Market Pipeline



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 The “Heat” Map — What Color Means

Heat ≠ Credit Spread
Heat = 1-Month Total Return in USD



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 No Double-Counting Rule

If country has...	Then we DON'T add...
Bond leg	Separate FX leg
Equity leg (US ETF)	Separate FX leg
Nothing	FX-only leg

💰 Three Cost Tiers — Same Trade, Different Bottom Line



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Example: \$50M Trade, 150 bps Vol

Component	Value
Base Discount	5%
Volume Discount	$\log_{10}(50) \times 5\% \approx 8.5\%$
Illiquidity Penalty	$(150-100)/1000 = 5\%$
Total Discount	8.5%
HF Pays	91.5% of retail markup



Straddle Fee — Empirical, Not Theoretical

Why? Fallen Angel spreads have mean-reversion ($AR(1) \approx -0.32$).

Shock day: 1-day $|\Delta| = 18$ bps $\rightarrow$ 5-day $|\Delta| = 16$ bps (whipsaw!)

Theoretical Bachelier: $\sqrt{5} \times 18 \approx 40$ bps **✗ 2.3x overprice**

Empirical fee: Rolling average of |T-day move| on **known** shock days (shifted by T, no look-ahead) **✓**



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Per-Country Legs (Live-Verified)

Region	Countries	Bond Yield	Equity ETF	Credit Index	Fallen Angel
US	1	FRED DGS10	—	BAMLC0A*	ANGEL
Euro Area	11	ECB LTIR 10Y	—	BAMLHE00EHYIOAS	EM1A
EM	32	FRED IRLTLT01*	EMB/CEMB/EMHY	BAMLHE00EHYIOAS	EM1A
EMEA	28	FRED IRLTLT01*	—	BAMLHE00EHYIOAS	GFA.I
LatAm	15	FRED IRLTLT01*	EMB/CEMB	BAMLHE00EHYIOAS	EM1A
Pacific	8	FRED IRLTLT01*	—	—	GFA.I
Dollarised	7	—	—	BAMLHE00EHYIOAS	—
CFA Franc	14	—	—	BAMLHE00EHYIOAS	—



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 Backtest — Walk-Forward, No Look-Ahead



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Live Results (2026-08 runs)

Leg	Hit Rate	IC	Sharpe	Sortino	MaxDD	OOS Sign Fit
CURVE	45.9%	-0.084	0.203	0.278	-5.4%	✅ +1 confirmed
COT	46.7%	—	0.116	—	-15%	❌ -1 flipped
IV-RV	<i>unlocking</i>	—	—	—	—	⌚ 3/20 days

Bugs You Fixed (Battle Scars)

Bug	Root Cause	Fix	Impact
FA premium overcharge	<code>any(g in name)</code> matched "B" in "BB"/"BBB"	Exact membership <code>name in DISTRESSED_GRADES</code>	BB retail +0.49→+1.15
Dealer markup 2x too high	Premium share = 1.0 (100% of IV-RV)	<code>DEALER_MARKUP_PREMIUM_SHARE = 0.3</code>	Avg markup 1.17→1.07
FRED CSV decommissioned	<code>fredgraph.csv</code> 404s everywhere	Port to JSON observations API	Web API live again
Atlas NaN → 500	Legacy NaN in <code>atlas.json</code>	<code>_clean_for_json</code> sanitizer	API stable
EM-CARRY dead	Lambda captured Ticker obj not symbol	Rebinding symbol string	Leg now VALIDATED
APPETITE 0 obs	Quarterly DR + <code>asfreq("ME")</code> = all-NaN	<code>resample("MS") + ffill(21)</code>	654 daily obs
Stub-IV overwrite	Degraded yfinance served 1e-5 IV	<code>IV_MIN_REAL = 0.02</code> gate	History survives
EM1ADE short history	36mo vs 60+ needed for joint VAR	Separate daily-frequency VAR	Now in Returns page

Your Contribution Checklist

- ☑ UI: Removed purple logo/brand from header
- ☑ UI: Pacific Time timestamp (auto PDT/PST)
- ☑ UI: Static `bundle.json` timestamp (not `new Date()`)
- ☑ Docs: Rewrote `CONTEXT.md` §3.5 – accessible methodology

- ☑ Features: ANGL straddle → US market Returns page
- ☑ Features: EM1A.DE straddle → Countries market Returns page
- ☑ Fix: Series alignment in `_trade_cost_basis()`
- ☑ Fix: Exact FA premium membership check
- ☑ Fix: Dealer markup premium share 0.3 (was 1.0)
- ☑ Fix: FA liquidity premium 1.05 (was 1.20)
- ☑ Fix: FRED CSV → JSON API (web layer)
- ☑ Fix: ECB URL path (data/YC/B.U2...)
- ☑ Fix: NaN sanitization in API
- ☑ Fix: `heatmap_db.py` `executemany` (psycopg3)
- ☑ Fix: FINRA breadth column matcher
- ☑ Fix: APPETITE composite DR resampling
- ☑ Fix: `IV_MIN_REAL = 0.02` stub-IV gate
- ☑ Test: 149 pure-logic unit tests passing

🔧 Tech Stack (What to Name-Drop)

Layer	Stack
Data	<code>pandas</code> , <code>numpy</code> , <code>yfinance</code> , <code>requests</code> (FRED), <code>fredapi</code>
Stats/ML	<code>statsmodels</code> (VAR/VECM, Johansen, GARCH), <code>xgboost</code> , <code>lightgbm</code> , <code>catboost</code> , <code>shap</code>
API	<code>fastapi</code> , <code>uvicorn</code> , <code>psycopg[binary]</code> , <code>redis</code>
DB	PostgreSQL + PostGIS (Docker), <code>sql/heatmap_schema.sql</code>
Frontend	Vite + ECharts, <code>web/</code> → Vercel serverless
Test	<code>pytest</code> (149 tests), <code>playwright</code> (browser automation)
Infra	<code>docker compose</code> (postgis + redis), <code>launchd</code> (IV accrual)

10 Soundbites for the Interview

1. "Public debt markets — gov bonds, corps, ETFs. Core question: where's compensation today?"
 2. "All free data: FRED, Yahoo, ECB, CFTC, World Bank. Each source gated, reported honestly."
 3. "Heat = 1-month USD total return per country. Bond + equity + credit + FA, averaged. No double-count."
 4. "Three P&L views: Gross (theory), HF Net (prime broker discount), Retail Net (full freight)."
 5. "Straddle fee is empirical — what market actually moved on similar days. Captures whipsaw."
 6. "Fallen angels: ANGL (US), EM1A.DE (EU UCITS), GFA.L (global UCITS). Not country-specific indices."
 7. "Board votes threshold-only: each leg needs walk-forward sign + OOS Sharpe > 0 to vote."
 8. "149 pure-logic tests, no network. LEDGER.md + CONTEXT.md = full audit trail."
 9. "Fixed FA premium overcharge (BB/BBB), dealer markup 2x, FRED CSV death, NaN 500s, APPETITE 0-obs."
 10. "No private credit, no synthetic data, no look-ahead. Everything live or honestly UNAVAILABLE."
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Cheat Sheet: Explain Like I'm 15

If They Ask...	You Say...
"What's a VAR?"	"Predicts each thing using its own past AND other things' past. Like forecasting temp AND humidity using yesterday's both."
"What's a z-score?"	"How many standard deviations from average."
"What's cointegration?"	"Two things that stay side-by-side long-term, even if they separate short-term. Like runners in a race."
"How's the straddle fee different?"	"Theoretical = formula assumes independent moves. Ours = what actually happened on similar days. Captures the whipsaw."
"Why three cost tiers?"	"HF gets volume discount (log scale) minus illiquidity penalty. Retail pays full markup. Gross = fantasy."
"What's the board vote logic?"	"No weights. Each leg is a z-score."
"What's walk-forward?"	"Train on past, test on future, roll forward. No peeking at test data during training."

Final Prep Checklist

- ☐ Read this doc front-to-back once
- ☐ Practice the 10 soundbites out loud
- ☐ Know the **3 bugs you're proudest of fixing**
- ☐ Be ready to say: *"Private credit? Removed entirely. No deal-level anything."*
- ☐ Have `INTERVIEW_PREP.md` open on screen for constants
- ☐ Remember: **It's OK to say "It's in config.py with the audit log"**

"The system answers: where is compensation in public credit markets right now, and what is the expected payoff of trading volatility in those segments?"

Generated for the interview. Good luck — you know this cold.